

Stock Sentiment Analysis using ML and NLP

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What is Sentiment Analysis?

Sentiment analysis is a natural language processing (NLP) technique that involves analyzing the sentiment expressed in a piece of text. It is widely used in various applications to understand the opinions, emotions, and attitudes of users towards a specific subject or topic.

In the context of the stock market, this project aims to predict bullish or bearish trends by analyzing news related to a particular stock, major geopolitical events, and industry-specific developments.

Overview

This project involves developing a machine learning model that uses news articles to gauge the overall sentiment associated with a specific stock. Additionally, technical parameters such as the open, high, and low prices of the stock over the past few days are incorporated as features to train the model.

Flow of Project

Web Scraping News Articles: Using the beautiful Soup Python library, news articles from various websites are scraped.

- Headlines and subheadings are collected from websites like the Economic Times, Money Control, Yahoo Finance, Live Mint, and MSN.

- Training the Model: The model is trained on news articles from the past 3-5 months for the following stocks:

 - HDFC BANK

 - Adani Enterprise

 - ITC

- Obtaining Technical Data: Using the yfinance Python library, data regarding the OHLC prices for the past 3 months for each of the three stocks is obtained.

- Calculating Sentiment Scores: The scraped data is analyzed using VADER to determine positivity, negativity, neutrality, and compound scores. Subjectivity and polarity are also assessed. Since the stock price-related data is predominantly factual, detecting nuanced sentiments like irony or sarcasm is relatively easy.

VADER calculates sentiment scores by looking up words in a sentiment lexicon, assigning sentiment scores to each word, summing these scores for the entire text, and then normalizing the sum to produce a final sentiment score.

Lexicon Lookup: VADER employs a pre-built lexicon (dictionary) containing words assigned sentiment scores.

A dataset is evaluated on a scale ranging from -4 to +4, where:

-4 signifies extremely negative sentiment,

-1 indicates negative sentiment,

0 represents neutral sentiment,

+1 signifies positive sentiment,

+4 denotes extremely positive sentiment.

Each word in the lexicon is linked to a polarity score (positive or negative) and an intensity score.

Polarity measures the positive or negative sentiment of a text. It's a float value within the range [-1.0, 1.0], where -1.0 indicates highly negative sentiment, 0.0 signifies neutral sentiment, and 1.0 indicates highly positive sentiment.

Subjectivity measures the subjective or opinionated nature of the text. It's also a float value within the range [0.0, 1.0], where 0.0 represents very objective (factual) content and 1.0 signifies very subjective (opinionated) content.

Model Training: Each stock was trained and tested using various algorithms, including Logistic Regression, LDA, Random Forest, Support Vector Machine, and Deep Neural

Networks. The model with the highest accuracy and precision was chosen. Additionally, the area under the ROC curve was analyzed, and the model that achieved the maximum area for both training and testing sets was ultimately used to further train the ensemble model.

The precision of each of these models was also given significant importance while selecting an appropriate algorithm for the ensemble model, considering the potential losses for investors due to false positives.

The results of training different models on each of the three stocks are as follows;

Classification Report for Logistic Regression:					Classification Report for SVM:				
	precision	recall	f1-score	support		precision	recall	f1-score	support
0	1.00	0.60	0.75	5	0	0.00	0.00	0.00	5
1	0.71	1.00	0.83	5	1	0.50	1.00	0.67	5
accuracy			0.80	10	accuracy			0.50	10
macro avg	0.86	0.80	0.79	10	macro avg	0.25	0.50	0.33	10
weighted avg	0.86	0.80	0.79	10	weighted avg	0.25	0.50	0.33	10

Classification Report for LDA:					Classification Report for Random Forest:				
	precision	recall	f1-score	support		precision	recall	f1-score	support
0	1.00	0.60	0.75	5	0	0.67	0.80	0.73	5
1	0.71	1.00	0.83	5	1	0.75	0.60	0.67	5
accuracy			0.80	10	accuracy			0.70	10
macro avg	0.86	0.80	0.79	10	macro avg	0.71	0.70	0.70	10
weighted avg	0.86	0.80	0.79	10	weighted avg	0.71	0.70	0.70	10

Classification Report for Deep Neural Network:				
	precision	recall	f1-score	support
0	0.50	1.00	0.67	5
1	0.00	0.00	0.00	5
accuracy			0.50	10
macro avg	0.25	0.50	0.33	10
weighted avg	0.25	0.50	0.33	10

Figure 3. Classification Report study for each trained model for HDFC

Classification Report for Logistic Regression:					Classification Report for SVM:				
	precision	recall	f1-score	support		precision	recall	f1-score	support
0	1.00	0.50	0.67	6	0	0.60	1.00	0.75	6
1	0.57	1.00	0.73	4	1	0.00	0.00	0.00	4
accuracy			0.70	10	accuracy			0.60	10
macro avg	0.79	0.75	0.70	10	macro avg	0.30	0.50	0.37	10
weighted avg	0.83	0.70	0.69	10	weighted avg	0.36	0.60	0.45	10

Classification Report for LDA:					Classification Report for Random Forest:				
	precision	recall	f1-score	support		precision	recall	f1-score	support
0	0.67	1.00	0.80	6	0	0.67	0.33	0.44	6
1	1.00	0.25	0.40	4	1	0.43	0.75	0.55	4
accuracy			0.70	10	accuracy			0.50	10
macro avg	0.83	0.62	0.60	10	macro avg	0.55	0.54	0.49	10
weighted avg	0.80	0.70	0.64	10	weighted avg	0.57	0.50	0.48	10

Classification Report for Deep Neural Network:				
	precision	recall	f1-score	support
0	0.00	0.00	0.00	6
1	0.40	1.00	0.57	4
accuracy			0.40	10
macro avg	0.20	0.50	0.29	10
weighted avg	0.16	0.40	0.23	10

Figure 5. Classification Report study for each trained model for Adani

Classification Report for Logistic Regression:					Classification Report for SVM:				
	precision	recall	f1-score	support		precision	recall	f1-score	support
0	1.00	1.00	1.00	7	0	0.70	1.00	0.82	7
1	1.00	1.00	1.00	3	1	0.00	0.00	0.00	3
accuracy			1.00	10	accuracy			0.70	10
macro avg	1.00	1.00	1.00	10	macro avg	0.35	0.50	0.41	10
weighted avg	1.00	1.00	1.00	10	weighted avg	0.49	0.70	0.58	10

Classification Report for LDA:					Classification Report for Random Forest:				
	precision	recall	f1-score	support		precision	recall	f1-score	support
0	1.00	0.86	0.92	7	0	0.78	1.00	0.88	7
1	0.75	1.00	0.86	3	1	1.00	0.33	0.50	3
accuracy			0.90	10	accuracy			0.80	10
macro avg	0.88	0.93	0.89	10	macro avg	0.89	0.67	0.69	10
weighted avg	0.93	0.90	0.90	10	weighted avg	0.84	0.80	0.76	10

Classification Report for Deep Neural Network:				
	precision	recall	f1-score	support
0	0.70	1.00	0.82	7
1	0.00	0.00	0.00	3
accuracy			0.70	10
macro avg	0.35	0.50	0.41	10
weighted avg	0.49	0.70	0.58	10

Figure 7. Classification Report study for each trained model for ITC

Training the ensemble model: After selecting the best model for each of the three stocks, we trained the ensemble model.

Since slight overfitting was evident in the individual models, the most suitable ensemble method to address this issue would be a Voting classifier.

A Voting classifier in ensemble learning combines multiple individual models (classifiers or regressors) and aggregates their predictions to arrive at a final prediction. This aggregation process effectively reduces the chances of overfitting.

Testing the model: After training the ensemble model, we used the model to predict on new testing data. Since the accuracy of the model highly depends on the amount of relevant news articles one feeds into it, it is advised that prior to making decisions based on market sentiment using this model, one should have enough news websites relevant to the stock(industry related news, geopolitical news- as they highly impact macroeconomic factors and therefore stock prices stock specific announcements-such as those related to dividends, bonuses ,rights issues or stock split/consolidation).

Finally, the model was tested by feeding news articles and technical parameters related to a new stock into the ensemble model.

Following are the parameters upon which the model was tested:

Sharpe Ratio

It measures the return of an investment compared to its risk.

A Sharpe ratio >1 is considered good. The risk free rate for this project was taken to be 0.

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Where:

R_p = The expected return of the portfolio (investment),

R_f = The risk-free rate of return,

$R_p - R_f$ = Portfolio's excess return

σ_p = The standard deviation of the portfolio's excess

Maximum Drawdown (MDD) represents the maximum loss incurred from the peak values of an investment to its lowest point before a new peak is reached. It is calculated as the difference between the peak value and the trough value, expressed as a percentage of the peak value.

Number of Trades: This parameter represents the total number of buy and sell transactions executed within a specific time period. It helps traders gain insights into market volatility and trends.

Win Ratio: This parameter measures the proportion of profitable trades or investments compared to the total number of trades or investments made over a given period. It provides an indication of a trader's success rate.

To calculate these parameters, you can use the yfinance library in Python. Simply provide the ticker symbol, start date, and end date, and define functions for each parameter. Note that there may be slight variations in the functions for short positions.

Results

The model was tested on new stocks:

Hindustan Unilever: Stock prices of HUL from 7th to 14th June 2024 fell significantly. As predicted by the ensemble model, going short on the stock would be advisable.

If we went short on HUL on 7th June and bought it on 14th then our returns would be as follows:

Date	Open	High	Low	Close	Number of Trades executed
05-06-2024	2533.550049	2723.95	2525	2602.75	5
06-06-2024	2600	2600	2518	2549.6	0
07-06-2024	2555.300049	2596.5	2525	2577.8	3
10-06-2024	2579	2593.65	2546	2565.35	3
11-06-2024	2566	2576.7	2551.3	2556.35	7
12-06-2024	2568	2568	2517	2528.7	9
13-06-2024	2487.949951	2505.8	2446.45	2487.4	3
14-06-2024	2473	2510.4	2470.4	2479.75	5

Overall Returns of the trade	Sharpe Ratio	22.4225
	Maximum Drawdown	0.0269
	Win Ratio	99.99%

Nvidia: Nvidia stock prices rose from 21st may to 18th June 2024, especially after the stock split announced on 7th June, thus our model should be predicting a bullish trend. The returns of the trade are as follows:

Date	Open	High	Low	Close	Number of Trades
21-05-2024	93.59899902	95.40000153	93.18	95.386	5
22-05-2024	95.45899963	96.01999664	93.249	94.95	0
23-05-2024	102.0279999	106.3199997	101.52	103.799	3
24-05-2024	104.4489975	106.4749985	103	106.469	3
28-05-2024	110.2440033	114.939003	109.883	113.901	7
29-05-2024	113.0500031	115.4919968	110.901	114.825	9
30-05-2024	114.6500015	115.8190002	109.663	110.5	3
31-05-2024	112.5199966	112.7170029	106.94	109.633	5
03-06-2024	113.6210022	115	112.003	115	2
04-06-2024	115.7160034	116.5999985	114.045	116.437	4
05-06-2024	118.3710022	122.4489975	117.468	122.44	7
06-06-2024	124.0479965	125.586998	118.32	120.998	6
07-06-2024	119.7699966	121.6920013	118.022	120.888	8
10-06-2024	120.3700027	123.0999985	117.01	121.79	8
11-06-2024	121.7699966	122.8700027	118.74	120.91	10
12-06-2024	123.0599976	126.8799973	122.57	125.2	1
13-06-2024	129.3899994	129.8000031	127.16	129.61	6
14-06-2024	129.9600067	132.8399963	128.32	131.88	7
17-06-2024	132.9900055	133.7299957	129.58	130.98	7

Overall returns of the trade	Sharpe Ratio	9.79
	Maximum Drawdown	5.44%
	Win Ratio	64.71%

Trading Strategy

Higher the accuracy for smaller time periods, the better the chances are for you to gain profits if you act according to the model's predictions.

If accuracy > 0.6 for the stock that you have predicted the labels for then it is advisable to rely on the model's predictions.

The more relevant the news articles are with respect to the time period and industry specific news, the better would be the model's accuracy (for instance, news regarding elections significantly impacted stock prices of Adani enterprise in the 2024 general elections).

Challenges Faced

During training, the model encountered difficulties when unique events like the stock price fall on June 4th for Adani Enterprise occurred. These events were outliers, making it challenging for the model to learn effectively. To address this, the model was retrained

by optimizing the class weights to account for the minority classes, such as stock price falls caused by sudden investor confidence loss.

Finding relevant news articles for web scraping proved to be a significant challenge. Various factors influence stock prices, and news related to a stock online is often limited to significant announcements and events. This limited data collection process made it time-consuming.

Way Forward

To maintain accuracy and relevance in dynamic market conditions, continuous monitoring and optimization of the models are essential. Incorporating user feedback and employing robust evaluation methods will further enhance the system's performance and usability over time.

Ultimately, a well-executed stock sentiment analysis project not only provides a better understanding of market sentiment but also empowers stakeholders with actionable insights to navigate the complexities of financial markets effectively.

We can also expand the scope of scraped sources to include social media platforms, such as using the Twitter API to extract tweets related to stocks.